Yuyu Liu

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EDUCATION BACKGROUND

State University of New York at Stony Brook, Ph.D. — **New York, USA**
Major: Computer Science — *Aug. 2025- present*

Southern University of Science and Technology (SUSTech), B.Sc. — **Guangdong, China**
Major: Mathematics — *Sep. 2020- Jul. 2024*

RESEARCH

- **Multivariate Time Series Root Cause Analysis via Granger Causal Discovery in Latent Space**. Yuyu Liu, Xin Wang, Chandra Reddy, Wesley M. Gifford, Tengfei Ma. *Preprint*.
- **HyperGuide: Hyperbolic Guidance for Efficient Multi-Step Reasoning in Large Language Models**. Yuyu Liu, Haotian Xu, Yanan He, Sarang Rajendra Patil, Mengjia Xu, Tengfei Ma. *Preprint*.
- **HypEHR: Hyperbolic Modeling of Electronic Health Records for Efficient Question Answering**. Yuyu Liu, Sarang Rajendra Patil, Mengjia Xu, Tengfei Ma. *Findings of ACL 2026*.

ONGOING RESEARCH

Agent-Managed Memory for Time-Series Foundation Models on Streaming Data — *Aug 2026 – present*
Joint work with IBM Research. Adds an agent-managed external memory to a frozen time-series foundation model so it can use history beyond its context window: the agent decides what to store, when to retrieve, and how to assemble it into the model's context.

Hyperbolic Memory for LLM Agents — *July 2026 – present*
Organizes an LLM agent's memory store as a hierarchy embedded in hyperbolic space and retrieves with hyperbolic distance, so queries match memories at the right level of abstraction rather than by flat cosine similarity.

WORK EXPERIENCE

Machine Learning Engineer Intern — **Meituan, Inc.**
Search Department, Retrieval Group — *Nov. 2024 - Apr. 2025*
- Built an end-to-end generative item retrieval channel for Meituan app search using Sentence-T5, adapting the dense retrieval framework of LIGER (Meta, 2025)
- Developed a user query category prediction model with a Qwen2.5-0.5B backbone to guide candidate generation, improving retrieval relevance and coverage for long-tail and ambiguous queries.

Machine Learning Engineer Intern — **Baidu, Inc.**
Digital Human Group — *Jan. 2024 – May 2024*
- Built a few-shot voice-cloning pipeline for live-streaming digital humans on GPT-SoVITS with LoRA fine-tuning, cloning a speaker's timbre from under 10 minutes of speech; trained in PyTorch on 8×A100 GPUs.
- Replaced the BERT text encoder in Bert-VITS2 with chinese-roberta-wwm-ext-large to improve naturalness of synthesized Chinese speech.
- Fixed a teeth-not-closing artifact in Unreal Engine 5 digital humans with a C++ post-processing routine on lip blendshape vectors.

SKILLS

Programming	Python, LaTeX
ML / LLM	PyTorch, Hugging Face Transformers and PEFT (LoRA), vLLM, DeepSpeed / FSDP, Faiss, geoopt (Riemannian optimization), scikit-learn
Data / Infra	Spark, Hive, Linux, Git, Docker, multi-GPU training, Weights & Biases
Research Areas	LLM agents and memory, LLM reasoning, generative and dense retrieval, time-series foundation models, hyperbolic representation learning